# Bug #138539 OPEN

## [HW-R0R2.0A.0Z.04]There are LAN SCAN COMPARE error during Reboot cycle test

Added by Kent.Lin 5 months ago. Updated 5 months ago.

| | | | |
|---|---|---|---|
| **Status:** | New | **Start date:** | |
| **Priority:** | Normal | **Due date:** | |
| **Assignee:** | Ernie.Hsu | | |
| **Target version:** | - | | |
| **Severity:** | Critical | **R&D Solution for fix:** | not fix |
| **Component:** | System | **Error Type:** | N/A |
| **HW Version:** | As [Test Configuartion] | **iDesign_Req_ID:** | |
| **FW Version:** | As [Test Configuartion] | **Assign_to_i:** | |
| **Issue Finder:** | DQA | **Customer :** | |
| **Fix_Reject_Counts:** | 0 | **Target Date:** | |
| **Solution(Root cause):** | New Issue | **Bug Create After MP:** | No |

**Description**

[Test Case]
Auto Reliability Test - System Reboot Test

[Expectation]
No LAN SCAN COMPARE errors should occur during reboot testing.

[Issue Description]
During the reboot cycle test, there were 69 instances of LAN SCAN COMPARE errors.

```
root@S621-efi:/mnt/x86UniTest_v1.0.4/log_04-Nov-2025_0833/R_105# grep -P 'Test: *[1-9][0-9]*'
Report/x86UniTest_Summary_Result.log
LAN  SCAN Compare Test:        69     /         100        /        100
PCIe SCAN Compare Test:        100     /         100        /        100
SDR  SCAN Compare Test:        1     /         100        /        100
SEC  AER  Check   Test:        1     /         100        /        100
SEC  MCE  Check   Test:        1     /         100        /        100
SEC  SEL  Check   Test:        2     /         100        /        100
LAN           R/W Test:        2     /         99        /        100
```

During the test, the device at BUS 10:00.0/1, NVIDIA CX-7, disappeared.

```
Read :
+LAN Number                 : 4
+LAN1 Vendor/Device ID      : 46:00.0 0200: 15b3:1021
+LAN1 Subsystem             : 15b3:0022
+LAN1 port name             : b46p0
+LAN1 firmware version      :  28.42.1000 (MT_0000000834)
+LAN1 mac address           :         ether 58:a2:e1:ca:76:5e  txqueuelen 1000  (Ethernet)
+LAN2 Vendor/Device ID      : 46:00.1 0200: 15b3:1021
+LAN2 Subsystem             : 15b3:0022
+LAN2 port name             : b46p1
+LAN2 firmware version      :  28.42.1000 (MT_0000000834)
+LAN2 mac address           :         ether 58:a2:e1:ca:76:5f  txqueuelen 1000  (Ethernet)
+LAN3 Name                  : Intel Corporation Ethernet Controller X710 for 10GBASE-T (rev 02)
+LAN3 Vendor/Device ID      : af:00.0 0200: 8086:15ff (rev 02)
+LAN3 root port LnkCap      : Speed 32GT/s, Width x4
+LAN3 root port LnkSta      : Speed 8GT/s, Width x4
+LAN3 LnkCap                : Speed 8GT/s, Width x4
+LAN3 LnkSta                : Speed 8GT/s, Width x4
+LAN3 Subsystem             : 8086:0000
+LAN3 port name             : bafp0
+LAN3 driver                : i40e 2.28.10
+LAN3 firmware version      :  9.30 0x8000f97a 1.3429.0
+LAN3 Speed                 :  10000Mb/s
+LAN3 Port                  :  Twisted Pair
+LAN3 mac address           :         ether 74:fe:48:b3:0d:9c  txqueuelen 1000  (Ethernet)
+LAN3 Product Name          : SKY-622G4-MBIO
+LAN3 Manufacture ID        : 13FE
+LAN3 Card Type             : N230
+LAN3 vendor firmware version: V9.30
+LAN3 Advantech FW Revision: V9.30.01
+
+LAN4 Name                  : Intel Corporation Ethernet Controller X710 for 10GBASE-T (rev 02)
+LAN4 Vendor/Device ID      : af:00.1 0200: 8086:15ff (rev 02)
+LAN4 root port LnkCap      : Speed 32GT/s, Width x4
+LAN4 root port LnkSta      : Speed 8GT/s, Width x4
+LAN4 LnkCap                : Speed 8GT/s, Width x4
+LAN4 LnkSta                : Speed 8GT/s, Width x4
+LAN4 Subsystem             : 8086:0000
+LAN4 port name             : bafp1
+LAN4 driver                : i40e 2.28.10
```

```
+LAN4 firmware version    :  9.30 0x8000f97a 1.3429.0
+LAN4 Speed               :  10000Mb/s
+LAN4 Port                :  Twisted Pair
+LAN4 mac address         :         ether 74:fe:48:b3:0d:ad  txqueuelen 1000  (Ethernet)
+LAN4 Product Name        : SKY-622G4-MBIO
+LAN4 Manufacture ID      : 13FE
+LAN4 Card Type           : N230
+LAN4 vendor firmware version: V9.30
+LAN4 Advantech FW Revision: V9.30.01
```

After the LAN SCAN COMPARE errors occurred during the reboot cycle test, the NVIDIA CX-7 device on LAN port 10 could not be detected.

[Additional Information]
During system disassembly and inspection of the NVIDIA CX-7 wiring, it was found that the cable connected to root port 10 was bent excessively, almost at a 90-degree angle.
round 32 occured the CX-7 HW error.

```
AER Error Check Test Start!

===PCI AER Initial Phase Check===
PCI device total:276
Initial Phase: 0 Device AER Error
PASS

===But Mask is set, [Notice, not an Error] ===
00:03.0 [8086:09a6] UnCorrectable Error: Warning for Unsupported Request Error
01:00.0 [8086:4944] UnCorrectable Error: Warning for Unsupported Request Error
03:00.0 [8086:2714] UnCorrectable Error: Warning for Unsupported Request Error
06:00.0 [8086:4944] UnCorrectable Error: Warning for Unsupported Request Error
08:00.0 [8086:2714] UnCorrectable Error: Warning for Unsupported Request Error
0b:00.0 [8086:4944] UnCorrectable Error: Warning for Unsupported Request Error
0d:00.0 [8086:2714] UnCorrectable Error: Warning for Unsupported Request Error
2c:00.0 [1a03:1150] UnCorrectable Error: Warning for Unsupported Request Error
2e:00.0 [1912:0014] UnCorrectable Error: Warning for Unsupported Request Error
2f:00.0 [14a4:2300] UnCorrectable Error: Warning for Unsupported Request Error
30:00.0 [14a4:2300] UnCorrectable Error: Warning for Unsupported Request Error
46:00.0 [15b3:1021] UnCorrectable Error: Warning for Unsupported Request Error
46:00.1 [15b3:1021] UnCorrectable Error: Warning for Unsupported Request Error
ae:00.0 [9005:028f] UnCorrectable Error: Warning for Unsupported Request Error
af:00.0 [8086:15ff] UnCorrectable Error: Warning for Unsupported Request Error
af:00.1 [8086:15ff] UnCorrectable Error: Warning for Unsupported Request Error

===PCI AER Runtime Phase Check===

Runtime Phase: 0 Device AER Error
PASS
```

```
===But Mask is set, [Notice, not an Error] ===

SEL Check Error : Failed
SEL ID:  0002
11/03/2025 21:43:55
Creator : bios (0x21)
SensorType Bit[7:0] =  Critical Interrupt
SEL ID:  0003
11/03/2025 21:43:55
Creator : bios (0x21)
SensorType Bit[7:0] =  Critical Interrupt
SEL ID:  0007
11/03/2025 21:45:01
Creator : bios (0x21)
SensorType Bit[7:0] =  Critical Interrupt
AER Error Check Test Finish!
--------------------------------------------------------------------------------
MCE Error Check Test Start!
FAIL

Dmesg Check Error: Failed
[ 35.475147] [Hardware Error]: event severity: fatal
[ 35.481087] [Hardware Error]: Error 0, type: fatal
[ 35.487066] [Hardware Error]: section_type: PCIe error
[ 35.493443] [Hardware Error]: version: 3.0
[ 35.498649] [Hardware Error]: command: 0xffff, status: 0xffff
[ 35.505718] [Hardware Error]: device_id: 0000:0f:02.0
[ 35.511975] [Hardware Error]: slot: 0
[ 35.516658] [Hardware Error]: secondary_bus: 0x00
[ 35.522493] [Hardware Error]: vendor_id: 0xffff, device_id: 0xffff
[ 35.529989] [Hardware Error]: class_code: ffffff
MCE Error Check Test Finish!
--------------------------------------------------------------------------------
Other Error Check Test Start!
PASS
No any Other Error
Other Error Check Test Finish!
--------------------------------------------------------------------------------
PCIe_Lane Error Check Test Start!
PASS
PCI device total:276, No any PCIe_Lane Error
PCIe_Lane Error Check Test Finish!
--------------------------------------------------------------------------------
SATA Error Check Test Start!
PASS
SATA Controller Total: 0, No any SATA Controller Error
```

```
SATA Device Total: 8, No any Device self-assessment Error
SATA Error Check Test Finish!
--------------------------------------------------------------------------------
SERR&PERR Error Check Test Start!
PASS
No any SERR&PERR Error
SERR&PERR Error Check Test Finish!
--------------------------------------------------------------------------------
SEL Error Check Test Start!
FAIL
SEL Check Error from fail_string: FAIL
   2 | 11/03/2025 | 21:43:55 | Critical Interrupt #0x03 | Bus Correctable error | Asserted
   3 | 11/03/2025 | 21:43:55 | Critical Interrupt #0x03 | Bus Fatal Error | Asserted
   5 | 11/03/2025 | 21:43:56 | Processor PROC_STATE | Uncorrectable machine check exception | Asserted
   7 | 11/03/2025 | 21:45:01 | Critical Interrupt #0x03 |  | Asserted
SEL Check Error: FAIL
SEL ID:  0004
11/03/2025 21:43:55
Creator : bios (0x21)
SensorType Bit[7:0] =  Processor
PROCESSOR CONFIGURATION_ERROR
CPU Socket Number : 0
Bank Type : IIO
Processor Error Type : Unknown
Error Severity (Correctable/Uncorrectable) : Fatal Error
Boot Type: Current Boot Error
SEL Check Error: FAIL
SEL ID:  0005
11/03/2025 21:43:56
Creator : bmc (0x20)
SensorType Bit[7:0] =  Processor
Machine Check Exception (Uncorrectable)
SEL Error Check Test Finish!
```

[Fail rate]
69/100

[Test Procedure]
1. Power on the system and enter the BIOS.
2. Run the OS reboot cycle test.
3. At the end of the test, observe the PCIe SCAN COMPARE error.

[Test Configuration]

| Configuration: | A |
|---|---|
| 1. MB rev. | SKY-621V4 MB A101-1 |
| -P/N: | 9692621V400 |
| -CPU: | Intel(R) Xeon(R) 6781P CPU @ 2.0GHz |
| -Memory: | Advantech DDR5 RDIMM 6400 MT/s<br>AQD-D5V32GR64-HB*16 |
| -BIOS Rev. | S621V406000X006<br>(EVALUATION ROM FOR V1.10)<br>(RMT_BIOS_Enable_DFX_X006) |
| -BMC Rev. | BL: 0.40/BMC: 1.03/<br>BMCONF : 0.05 |
| -FPGA Rev. | v0.01 |
| -NVRAM Rev. | V4.00 |
| 2. Other Board Rev. | SKY-621V4 MB A101-1<br>SKY-621V4 PDB A101-1<br>SKY-621V4 MBIO A101-1<br>SKY-8132S MBIO A101-1<br>ASMB610 FLB A101-1<br>622G4 IO Board A101-1 |
| 3. Power Supply: | FSP2700-29HM*2 |

**Other Equipment:**

1. M.2 NVMe: Lite-On CA3-8D512 512G*2
2. 2.5" SSD (SATA): Micron_1300_MTFDDAK512TDL*8

3. GPU Card: NVIDIA Corporation AD102GL [L40S] (Gen4/x16) *4
4. RAID Card: Adaptec SmartRAID 3102-8i *1
5. Network Card: Intel Corporation Ethernet Network Adapter X710-TL*1
6. Network Card: Mellanox Technologies MT2910 Family ConnectX-7*2

**Test OS:**
OS: 24.04.3 LTS | Linux kernal: 6.8.0-86-generic
ADV_BSP_FILE= GID-Ubuntu24.04_1.61.063_2025.10.21.img

**Files**

| | |
|---|---|
| clipboard-202511071853-rpmun.png (55.3 KB) | Kent.Lin, 2025-11-07 06:53 PM |
| clipboard-202511071857-dubea.png (15.6 KB) | Kent.Lin, 2025-11-07 06:57 PM |
| clipboard-202511071859-ispbb.png (84 KB) | Kent.Lin, 2025-11-07 06:59 PM |
| clipboard-202511111459-6ehno.png (37.6 KB) | Kent.Lin, 2025-11-11 02:59 PM |

**Subtasks**

**Related issues**

**History** | **Property changes**

Updated by Kent.Lin 5 months ago

- **File** clipboard-202511111459-6ehno.png added
- **Description** updated (diff)

Updated by Kent.Lin 5 months ago

- **Story points** set to *0.0*
- **Tracker** changed from *EE_Task* to *Bug*
- **Description** updated (diff)
- **Assignee** set to *Ernie.Hsu*
- **HW Version** changed from *A101-1* to *As [Test Configuartion]*
- **Component** set to *(none)*
- **FW Version** set to *As [Test Configuartion]*

- **Issue Finder** set to *DQA*
- **Fix_Reject_Counts** set to *0*
- **Solution(Root cause)** set to *New Issue*
- **R&D Solution for fix** set to *not fix*
- **Bug Create After MP** set to *No*

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Updated by Kent.Lin 5 months ago

- **Severity** changed from *Low/Cosmetically* to *Critical*
- **Component** changed from *(none)* to *System*

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Updated by Kent.Lin 5 months ago

- **Description** updated (diff)